Tutorial 2: steps to run

Just the commands, in order. Explanations and questions are in `2_TUTORIAL.pdf`. Type `namd3` instead of `namd2` if that is what you installed. Change `+p4` to your number of CPU cores: `nproc` (Ubuntu) or `sysctl -n hw.ncpu` (macOS). Commands are the same on Ubuntu and macOS unless marked.

Before class (once)

Full details and fixes: `1_BEFORE_CLASS_Install.pdf`. First download the NAMD **multicore** build for your OS (free registration) into `~/Downloads`.

Ubuntu

```
cd ~/Downloads
tar -xzf NAMD_*_Linux-x86_64-multicore.tar.gz
mv NAMD_*_Linux-x86_64-multicore ~/namd
echo 'export PATH=$HOME/namd:$PATH' >> ~/.bashrc
source ~/.bashrc
```

macOS

```
cd ~/Downloads
tar -xzf NAMD_*_MacOS*-multicore.tar.gz
mv NAMD_*_MacOS*-multicore ~/namd
xattr -dr com.apple.quarantine ~/namd
echo 'export PATH=$HOME/namd:$PATH' >> ~/.zshrc
source ~/.zshrc
```

Both (needs conda; install Miniconda first if `conda --version` fails)

```
cd ~/Downloads/CIS1_Tutorial2_NVE        # wherever you unzipped it
conda env create -f environment.yml --solver=libmamba   # creates cis1-md (once, ~2 min)
conda activate cis1-md
python check_install.py                  # last line must say: ALL GOOD
```

Step 0. Open the folder · terminal 1

```
cd ~/Downloads/CIS1_Tutorial2_NVE
conda activate cis1-md
ls inputs
head -3 inputs/trpcage_equil.rst7        # title line, atom count (6215), then x y z
```

Step 1. Run NAMD (5–15 min) · terminal 1

```
cd namd
namd2 +p4 nve.namd > out/nve.log         # prints nothing; output goes to the log
```

While it runs, watch the energies in **terminal 2**:

```
cd ~/Downloads/CIS1_Tutorial2_NVE/namd
grep ^ETITLE out/nve.log | head -1       # column names
grep ^ENERGY out/nve.log | tail -3       # latest energies (kcal/mol)
```

Open `nve.namd` in a text editor and find `parmfile`, `ambercoor`, `cellBasisVector`, `cutoff`, `PME`, `timestep`, `rigidBonds`, `temperature` and the two `run` lines. Check that there is **no** `langevin` or `langevinPiston` line: that is what makes it NVE.

NAMD is done when this shows `End of program`:

```
tail -3 out/nve.log
```

Step 2. Run OpenMM (2–6 min) · terminal 2

```
cd ~/Downloads/CIS1_Tutorial2_NVE/openmm
conda activate cis1-md
python nve.py 1000                       # 1000 steps x 2 fs = 2 ps
head -3 nve.csv                          # step, time (ps), PE, KE, total (kJ/mol), T (K)
```

While it runs, open `nve.py` and match each block to `nve.namd` (table in `4_Cheatsheet_NAMD_vs_OpenMM.pdf`).

Step 3. Plot (after both runs have finished)

```
cd ~/Downloads/CIS1_Tutorial2_NVE/analysis
python plot_nve.py                       # your NAMD run + your short OpenMM run
xdg-open nve_energies.png                # Ubuntu
open nve_energies.png                    # macOS
```

Ignore the drift number of your 2 ps OpenMM run: it is too short. Now plot the full 60 ps reference runs:

```
python plot_nve.py ../reference_results/namd_nve.log ../reference_results/openmm_nve.csv
xdg-open nve_energies.png                # macOS: open nve_energies.png
```

Step 4. Answer the questions

See *Step 4. Questions* in `2_TUTORIAL.pdf`.

Step 5. Exercise: change the timestep (remaining time or homework)

```
cd ~/Downloads/CIS1_Tutorial2_NVE/namd
cp nve.namd nve_dt1.namd
cp nve.namd nve_dt4.namd
```

Edit the copies:

File	`timestep`	first `run`	second `run`	`outputEnergies`	`outputName`
`nve_dt1.namd`	1.0	10000	50000	200	`out/nve_dt1`
`nve_dt4.namd`	4.0	2500	12500	50	`out/nve_dt4`

```
namd2 +p4 nve_dt1.namd > out/nve_dt1.log
namd2 +p4 nve_dt4.namd > out/nve_dt4.log
cd ../analysis
python plot_nve.py ../namd/out/nve_dt1.log -     # '-' = NAMD only
python plot_nve.py ../namd/out/nve_dt4.log -
```

Compare `std(Etot)` and drift with your 2 fs run.

Optional. Watch the movie in VMD

```
cd ~/Downloads/CIS1_Tutorial2_NVE/namd
vmd ../inputs/trpcage_wat.prmtop -parm7 -dcd out/nve.dcd     # Ubuntu
```

macOS: start VMD from Applications, *File → New Molecule*, load `inputs/trpcage_wat.prmtop` as type **parm7**, then load `namd/out/nve.dcd` into the same molecule.

If something goes wrong

Problem	Fix
`EnvironmentNameNotFound: ... cis1-md`	`conda env create -f environment.yml` (in the tutorial folder), then `conda activate cis1-md`
`namd2: command not found`	Find NAMD: `ls -d ~/namd ~/NAMD_*`. Ubuntu: `echo 'export PATH=$HOME/namd:$PATH' >> ~/.bashrc; source ~/.bashrc`. macOS: same with `~/.zshrc`. Replace `namd` with your folder name
macOS: *developer cannot be verified*	`xattr -dr com.apple.quarantine ~/namd`
`NoWritablePkgsDirError`	`CONDA_PKGS_DIRS=$HOME/miniconda3/pkgs conda env create -f environment.yml`
`out/nve.log: No such file`	run `mkdir -p out` inside the `namd` folder
`No module named 'openmm'`	run `conda activate cis1-md` first
NAMD says `FATAL ERROR`	read the next line; if you edited `nve.namd`, copy a fresh one from the zip
plot says `No ENERGY: lines`	NAMD has not finished or crashed: `tail out/nve.log`
OpenMM very slow	use fewer steps: `python nve.py 500`